Academic Thesis Title

Thesis subtitle (optional)

Student:

Student Name

Supervisor:

Supervisor Name

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Program:

Program / Master’s Degree / Course

Place and Date

Acknowledgements

Write acknowledgements here.

Abstract

Write the thesis abstract here.

Keywords: keyword 1, keyword 2, keyword 3, keyword 4.

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Chapter 1

Introduction

1.1 Context or Scope

Describe the context and scope of the project.

1.2 Additional Section

Add the required subsection content here.

1.3 Justification

Explain why this project is important.

1.4 Objectives

1.4.1 General Objective

State the overall objective.

1.4.2 Specific Objectives

- Specific objective 1.
- Specific objective 2.
- Specific objective 3.

1.5 Project Limits

Define the boundaries and limitations of the project.

Chapter 2

Project Background

2.1 Sector Analysis

Analyze the domain and current landscape.

2.2 Current Tools

Describe currently used tools and platforms.

2.3 Current Techniques

Describe current methods and technical approaches.

2.4 Conclusions / Summary

Summarize key findings from the background analysis.

Chapter 3

Project Hypothesis and Proposed Method

3.1 Research Question

This thesis investigates a practical question in applied machine learning: *is SHAP-based feature selection worth its additional complexity and computational cost when compared with simpler feature-importance-based alternatives?* The comparison is made under a unified evaluation framework that jointly considers predictive quality, runtime cost, feature-set stability, and qualitative interpretability evidence.

The motivation is straightforward. SHAP-derived rankings are often perceived as more principled because they are connected to game-theoretic attribution, but in production settings the selection method must also be justified in terms of reproducibility and efficiency. This work treats feature selection as an engineering decision problem, not only a statistical ranking problem.

3.2 Multi-Stage Hypothesis Structure

The thesis is organized around a staged hypothesis design.

3.2.1 H1: Screening hypothesis (Experiment 1)

In a broad benchmark across multiple datasets, models, and selection techniques, SHAP-based and tree-importance-based selection are expected to be among the most competitive methods. This stage is exploratory and is used to identify promising candidates for deeper analysis.

3.2.2 H2: Controlled performance hypothesis (Unified Experiment 2)

When the model family and protocol are fixed, SHAP-based selection is not expected to produce consistent predictive gains over simpler feature-importance alternatives. In other words, performance parity is expected to be common.

3.2.3 H3: Computational cost hypothesis (Unified Experiment 2)

Even when predictive performance is similar, SHAP-oriented pipelines are expected to incur higher computational overhead, especially in selection-heavy variants.

3.2.4 H4: Redundant-subspace hypothesis (Deep-dive cases)

Near-identical predictive performance can emerge from substantially different selected feature subsets. This implies that, for these datasets, multiple feature subspaces can support similar model quality.

3.3 Experimental Design Overview

The empirical design has two phases.

3.3.1 Phase I: Experiment 1 (broad benchmark)

Experiment 1 compares three regression models (Ridge, Extra Trees, XGBoost), three datasets (Bike Sharing, Communities and Crime, Superconductor), and four feature selection techniques (mutual information, recursive feature elimination, SHAP-based ranking, and tree-based importance ranking). The goal is not final model selection, but candidate screening under heterogeneous conditions.

3.3.2 Phase II: Unified Experiment 2 (focused validation)

A single, controlled protocol is then applied to two datasets:

- **Experiment 2A:** Superconductor dataset.
- **Experiment 2B:** Allstate dataset (ARFF source with categorical support enabled).

Although implemented as separate scripts during development, both runs represent the same experiment at the thesis level: same model family, same strategy definitions, same cross-validation/repetition logic, and same output schema.

3.4 Materials and Computational Setup

3.4.1 Data

The thesis uses five datasets in total across phases. Experiment 1 uses Bike Sharing, Communities and Crime, and Superconductor. Unified Experiment 2 uses Superconductor (2A) and Allstate (2B), where 2B introduces high-cardinality categorical variables and a different signal-to-noise regime.

3.4.2 Model family

Unified Experiment 2 fixes the predictor to XGBoost to remove model-choice confounding and isolate the effect of feature selection. For Allstate, categorical handling is enabled directly in the model pipeline.

3.4.3 Selection strategies in Unified Experiment 2

The following strategies are compared:

- `baseline`: no feature dropping at `k=1.0`.
- `native_fi`: feature selection from native model importance ranking.
- `custom_shap_rfe`: SHAP-guided recursive elimination loop.
- `hybrid_fi_custom_shap_rfe`: FI pre-filtering followed by SHAP-guided elimination.

Retention ratios are evaluated over a grid of `k` values, allowing a cost-performance tradeoff analysis under varying feature budgets.

3.5 Evaluation Criteria

Evaluation is multi-dimensional:

- **Predictive quality:** RMSE, MAE, and MAPE.
- **Computational efficiency:** selection, training, prediction, and total runtime.
- **Stability and structural behavior:** consistency and overlap of selected feature subsets across folds and seeds.

- **Interpretive evidence:** targeted case studies comparing similarly performing runs with distinct selected subsets.

This design prevents over-reliance on a single metric and supports a practical conclusion about methodological utility.

3.6 Proposed Contributions

The thesis contributes in three ways:

- A staged protocol that links broad benchmarking to controlled replication.
- A unified Experiment 2 framing that treats cross-dataset replication as a first-class requirement.
- A case-based analysis showing how feature subset identity and predictive parity can coexist, clarifying when SHAP selection is an interpretability choice rather than a pure performance choice.

3.7 Validity Considerations

Three boundaries should be acknowledged. First, conclusions are grounded in the studied datasets and model class, not universal across all domains. Second, practical equivalence in predictive metrics is interpreted in the context of repeated folds and seeds, not from single-run differences. Third, runtime outcomes are implementation- and hardware-sensitive, though the relative overhead patterns remain informative for engineering decisions.

Chapter 4

Methodology and Results

4.1 Methodological Roadmap

The empirical work follows a two-stage logic. Stage 1 (Experiment 1) is a broad screening study designed to identify competitive feature selection candidates under diverse conditions. Stage 2 (Unified Experiment 2) is a controlled study that reuses one protocol on two datasets, denoted as Experiment 2A (Superconductor) and Experiment 2B (Allstate).

This structure is intentional: broad evidence is used to select promising directions, while controlled replication is used to validate whether those directions remain advantageous when confounding factors are reduced.

Table 4.1: Experiment 1 winners by dataset-model pair (lower RMSE is better).

Dataset	Model	Strategy	RMSE	Selection time (s)	Total time (s)
Bike Sharing	extratrees	tree	58.6830	1.6236	3.1385
Bike Sharing	ridge	rfe	143.4365	0.0537	0.0568
Bike Sharing	xgboost	tree	57.5420	1.9285	2.9359
Communities and Crime	extratrees	tree	0.1344	1.5295	2.9004
Communities and Crime	ridge	rfe	0.1367	0.7994	0.8358
Communities and Crime	xgboost	mi	0.1384	1.1298	2.2558
Superconductor	extratrees	rfe	9.2425	1.0803	6.7431
Superconductor	ridge	rfe	19.0345	1.6656	1.6906
Superconductor	xgboost	shap	10.2085	22.8623	24.1957

4.2 Stage 1: Experiment 1 Screening

Experiment 1 evaluates three datasets, three models, and four feature selection methods. The output of this phase is used as a decision filter for later experiments.

The winner table shows that no single strategy dominates every dataset-model pair. However, tree-based and SHAP-based methods remain competitive in the benchmark and motivate focused follow-up under stricter control. This result is important for the narrative of the thesis: a method can appear promising in broad comparisons yet still fail to deliver a robust cost-adjusted advantage in a controlled setting.

4.3 Stage 2: Unified Experiment 2 Design (2A + 2B)

Unified Experiment 2 standardizes model family, validation protocol, and strategy definitions, then applies the same procedure to two datasets with different feature characteristics.

4.3.1 Protocol

For each dataset instance, repeated folds and seeds are executed across a grid of feature-retention ratios. At each run unit, the process is selection $\rightarrow$ model fit $\rightarrow$ prediction $\rightarrow$ metric and runtime logging. All outputs are persisted in run-isolated directories and can be resumed safely.

4.3.2 Compared strategies

The comparison includes `native_fi`, `custom_shap_rfe`, and `hybrid_fi_custom_shap_rfe` with `baseline` represented by `k=1.0`. This setup isolates whether SHAP-centric selection improves quality enough to justify additional computational cost.

4.4 Aggregate Results for Unified Experiment 2

4.4.1 Predictive behavior across feature budgets

Figure 4.1 summarizes RMSE trends across `k` for both dataset instances. Performance differences among strategies are generally modest relative to absolute target scale, especially at medium and high retention levels.

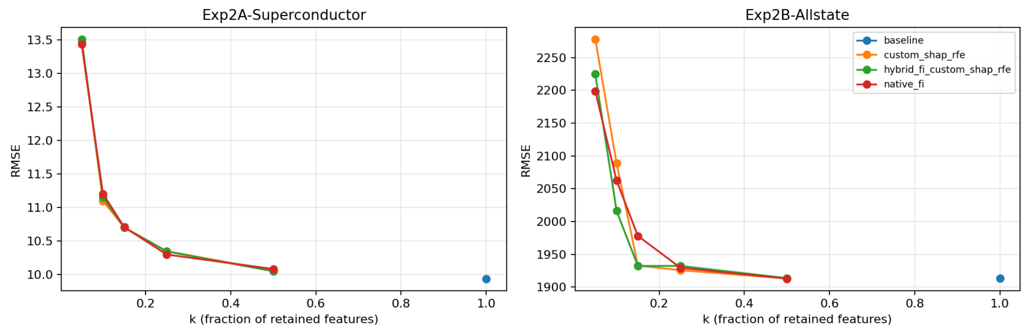

Figure 4.1: Unified Experiment 2: RMSE versus retained-feature ratio (k) for Experiment 2A and 2B.

4.4.2 Runtime behavior and compute asymmetry

Figures 4.2 and 4.3 show the most consistent result in this thesis: SHAP-heavy hybrid selection introduces a large runtime premium, while predictive differences remain comparatively small.

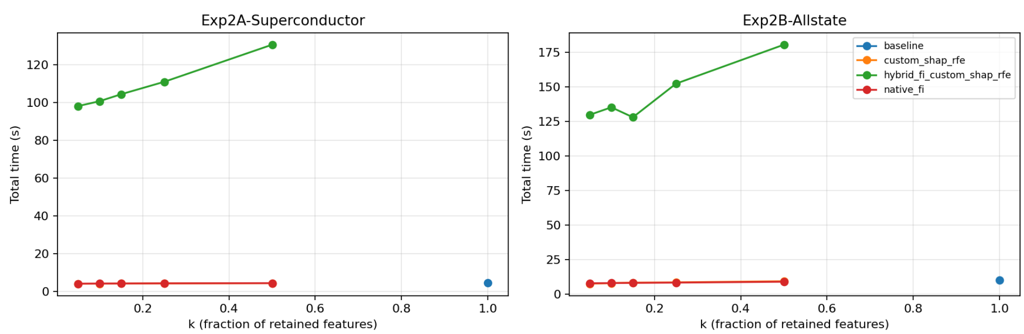

Figure 4.2: Unified Experiment 2: total runtime versus k for Experiment 2A and 2B.

Table 4.2 provides representative numeric snapshots at `k=0.10`, `k=0.50`, and `k=1.00`. At `k=0.10`, both datasets already show the same qualitative pat-

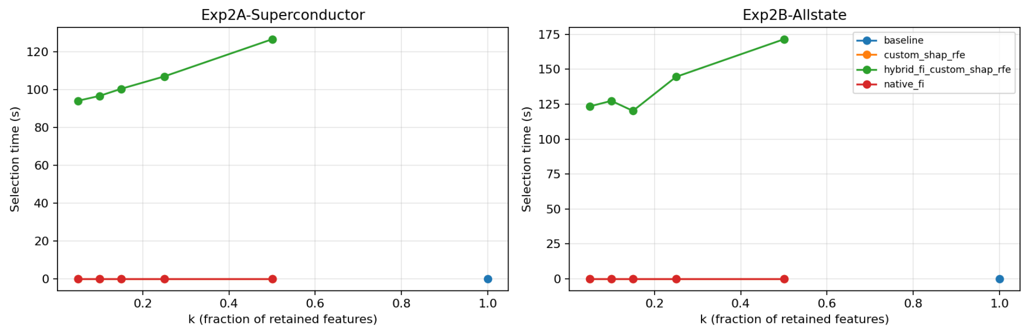

Figure 4.3: Unified Experiment 2: selection runtime versus k for Experiment 2A and 2B.

tern: close predictive metrics across strategies, but a major compute gap for hybrid SHAP-oriented selection. At `k=1.00`, baseline behavior confirms the expected full-feature reference.

4.5 Cross-Dataset Synthesis (2A and 2B)

Taken together, Experiment 2A and 2B support a common conclusion: feature selection strategy changes often produce second-order differences in predictive quality compared with first-order differences in computational cost. The pattern is not absolute in every fold-seed instance, but it is stable enough at aggregate level to inform engineering choice.

This does not imply SHAP-based selection is ineffective. Instead, it suggests that in these settings SHAP's strongest value is interpretive and diagnostic, while pure predictive gains are limited.

Table 4.2: Unified Experiment 2 summary at representative k levels.

Dataset instance	k	Strategy	RMSE	MAE	Selection ti
Exp2A-Superconductor	0.1000	custom_shap_rfe	11.0985	7.0121	0
Exp2A-Superconductor	0.1000	hybrid_fi_custom_shap_rfe	11.1505	7.0530	96
Exp2A-Superconductor	0.1000	native_fi	11.2040	7.1085	0
Exp2A-Superconductor	0.5000	custom_shap_rfe	10.0555	6.2661	0
Exp2A-Superconductor	0.5000	hybrid_fi_custom_shap_rfe	10.0453	6.2519	126
Exp2A-Superconductor	0.5000	native_fi	10.0794	6.2717	0
Exp2A-Superconductor	1.0000	baseline	9.9358	6.1597	0
Exp2B-Allstate	0.1000	custom_shap_rfe	2088.8803	1308.9147	0
Exp2B-Allstate	0.1000	hybrid_fi_custom_shap_rfe	2016.1172	1292.5887	127
Exp2B-Allstate	0.1000	native_fi	2062.3931	1338.7326	0
Exp2B-Allstate	0.5000	custom_shap_rfe	1913.0268	1195.8560	0
Exp2B-Allstate	0.5000	hybrid_fi_custom_shap_rfe	1913.7163	1195.7562	171
Exp2B-Allstate	0.5000	native_fi	1912.9177	1195.6843	0
Exp2B-Allstate	1.0000	baseline	1913.1240	1195.6906	0

Table 4.3: Showcase cases with near-identical RMSE and divergent feature subsets.

Dataset instance	k	Fold	Seed	Strategy A	Strategy B	Abs. RMSE diff
Exp2A-Superconductor	0.100000	6	123	native_fi	custom_shap_rfe	0.002737
Exp2B-Allstate	0.100000	6	999	native_fi	custom_shap_rfe	0.126220
Exp2A-Superconductor	0.150000	6	123	native_fi	custom_shap_rfe	0.001648

4.6 Deep-Dive Case Studies: Similar Accuracy, Different Features

Aggregate metrics can hide structural differences in selected subsets. To expose that behavior, three showcase cases were selected with two criteria: near-identical RMSE and non-trivial feature-set divergence.

In the selected cases, relative RMSE differences are extremely small, while Jaccard overlap remains moderate-to-low and symmetric feature differences are non-negligible. This supports the redundant-subspace hypothesis: different feature subsets can map to effectively the same predictive quality.

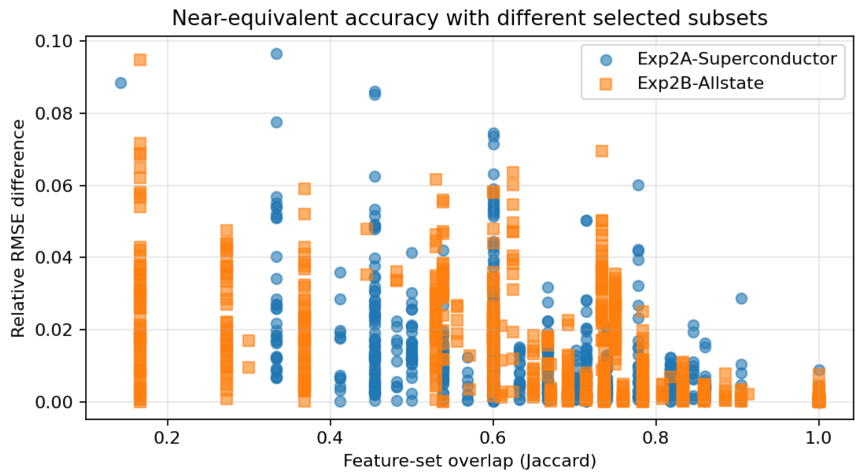

Figure 4.4: Candidate-pair landscape: relative RMSE difference versus feature-set overlap (Jaccard).

The scatter in Figure 4.4 reinforces the point visually: low error deltas can occur even when overlap is far from complete. From a practical perspective, this weakens the assumption that one "correct" subset must emerge from the selection process.

4.7 Interpretation

The combined evidence suggests an efficiency-utility asymmetry:

- Predictive quality is often similar across `native_fi`, `custom_shap_rfe`, and hybrid variants.
- SHAP-heavy selection paths can substantially increase runtime.
- Feature subset identity is not unique, even when predictive outcomes are nearly identical.

Therefore, selecting SHAP-based feature elimination should be justified primarily by interpretability requirements, auditability needs, or model-understanding goals, rather than by an expectation of consistent metric improvements.

4.8 Threats to Validity

Several limitations are relevant. First, conclusions are tied to the chosen datasets and XGBoost-centered protocol. Second, runtime values depend on implementation details and available hardware, even though relative trends are informative. Third, "practical equivalence" in predictive quality is contextual and should be interpreted with domain-specific tolerances. Finally, deep-dive case studies are explanatory complements to aggregate statistics, not replacements for them.

4.9 Chapter Conclusion

This chapter showed how broad screening (Experiment 1) and controlled replication (Unified Experiment 2A/2B) interact to answer the thesis question. The evidence indicates that SHAP-based selection is a strong interpretive tool but not a consistent cost-effective winner in predictive performance. The most robust empirical signal is the recurring gap between marginal quality differences and substantial computational overhead.

Chapter 5

Conclusion and Future Development

5.1 Conclusion

Provide the final conclusions of the project.

5.2 Future Work to Be Developed

Describe future improvements and possible continuation lines.

Bibliography

Bibliography